

NATIONAL UNIVERSITY OF SINGAPORE

FINAL ASSESSMENT FOR CS2105 – INTRODUCTION TO COMPUTER NETWORKS (Semester 2: AY2016/2017)

Time allowed: 2 hours

INSTRUCTIONS TO CANDIDATES

1. This assessment paper contains **SIX** questions and comprises **FOURTEEN** printed pages.
2. This is an **OPEN BOOK** assessment. The maximum possible score is **55 marks**.
3. Calculators are allowed, but not laptops, PDAs, or other electronic devices.
4. Fill in your student number clearly below.

STUDENT NO:

--	--	--	--	--	--	--	--	--

For examiners' use only							
Question	Q1	Q2	Q3	Q4	Q5	Q6	Total
Max	11	7	7	12	9	9	55
Score							

For Q2-Q6, write your answers within the space provided in each question.

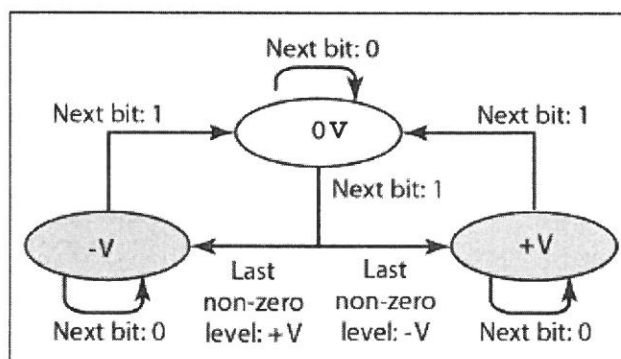
Q2.

[Total: 7 marks]

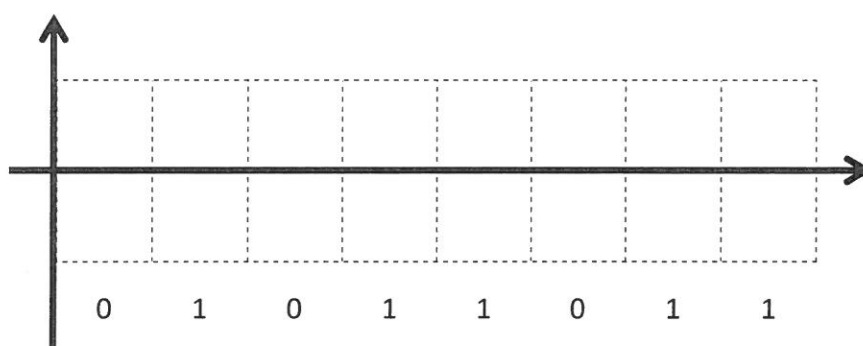
- (a) [2 marks] Consider a noisy channel with a Shannon capacity of 50 Kbps and a bandwidth of 10 KHz. What is the SNR of this channel?

- (b) [2 marks] A CSMA/CD network spans 2000 meters. The minimum frame size required for this network is 40 bits. Suppose the speed of electromagnetic wave is 2×10^8 meters/s, what is the maximum data rate of this network?

- (c) [3 marks] MLT-3 (Multi-level transmit, three levels) is a digital encoding method that uses three voltage levels. The encoding rules are shown in the transition diagram on the right. Draw the time domain graph below for the bit pattern 01011011.



You may assume that the state prior to the bit pattern is "state 0" and the last non-zero level is +V.



Q3. Keep your answers accurate and concise.

[Total: 7 marks]

- (a) **[2 marks]** Name an application that uses UDP service and another application that uses TCP service. Describe the reason they choose the respective transport service.

- (b) **[2 marks]** Why is it necessary for an IP datagram to carry both the source and the destination IP addresses?

(c) **[3 marks]** To preserve message confidentiality and authenticity, the following information is contained in a secured message sent from Alice to Bob.

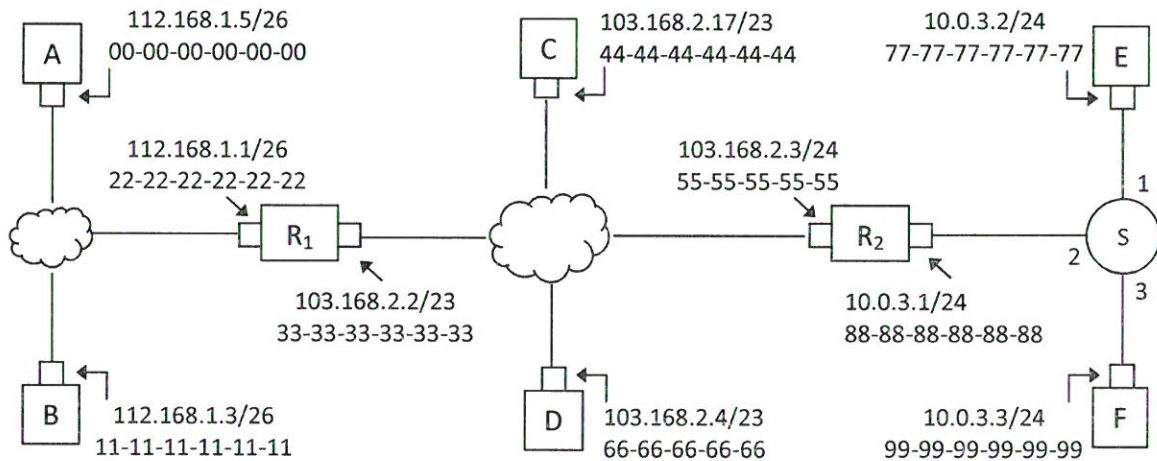
- Encrypted hash of the message
- Encrypted message
- Encrypted session key

Briefly describe the purpose of each piece of information and the key used in generating that information.

Q4.

[Total: 12 marks]

The diagram below shows three LANs interconnected by two routers (R_1 & R_2), with the IP addresses and MAC addresses of various hosts ($A...F$) and routers given. In particular, R_2 is an NAT-enabled router. S is a switch with 3 interfaces labelled as 1, 2 and 3. Suppose every host and router is aware of the IP address of every other host and router, where applicable.



- (a) [3 marks] Consider sending an IP datagram from Host E to Host F . Suppose all ARP tables, switch table and forwarding tables are up to date. Outline the main steps and sequence of events/actions required at respective devices (i.e. host/router/switch) to transmit this datagram. Keep your answer accurate and concise.

- (b) [4 marks] Consider sending an IP datagram from Host F to Host B . Suppose all ARP tables are empty but all other tables are up to date. Outline the main steps and sequence of events/actions required at respective devices to transmit this datagram. Keep your answer accurate and concise.

(c) [3 marks] Suppose the switch table of S is initially empty but all the other tables are up to date. The following events happen in sequence:

- i. F sends an IP datagram to A
- ii. E pings F
- iii. A replies with an IP datagram to F

Show the state of the switch table after each of the above events in the format: <MAC address, Interface>. You may ignore TTL field.

Event	Switch table after the event
F sends an IP datagram to A	
E pings F	
A replies with an IP datagram to F	

(d) [2 marks] Suppose you are the network administrator of the LAN consisting of E , F and R_2 . Now you launch a Web server on host E . How can you configure your network so that external hosts such as B and C can access your Web server? Keep your answer succinct.

Q5.

[Total: 9 marks]

(a) A sliding window protocol is used for transmission. What is the sender utilization given the following parameters of transmission?

- i. [2 marks] RTT = 24 milliseconds, link rate = 1 Mbps, packet size = 125 bytes, window size = 10

- ii. [2 marks] RTT = 10 milliseconds, link rate = 100 Mbps, packet size = 1300 bytes, window size = 100

(b) [2 marks] To make the TCP initial sequence number a random number, most systems start the counter at 1 during bootstrap and increment the counter by 64,000 every 0.5 s. How long (in seconds) does it take for the counter to wrap around?

(c) [3 marks] Is it possible for a TCP sender to receive an ACK for a packet that falls outside of its current send window? Answer "Yes" or "No" and briefly justify your answer.

Q6.

[Total: 9 marks]

Consider a path from node 0, through nodes 1, 2, ..., till node $K+1$. The links, from node i to node $i+1$, for $i = 0, 1, \dots, K$ each has the same link rate r bits/s and propagation delay p seconds. Every node, from node 1 through node K , is a store-and-forward device; it begins forwarding a received packet μ seconds after it has finished receiving it.

- (a) [3 marks] Suppose node 0 sends a message of m bits to node $K+1$ without message segmentation. What is the end-to-end delay D for transmission (i.e. the duration from when the first bit of the message leaves node 0 to when the last bit of the message arrives at node $K + 1$)?

$$D =$$

- (b) [4 marks] Suppose node 0 sends a message of m bits to node $K+1$ as a series of consecutive packets. Each packet contains h header bits and l data bits. Assume m is divisible by l and $m \gg (h + l)$. What is the end-to-end delay D for transmission (i.e. the duration from when the first bit of the first packet leaves node 0 to when the last bit of the last packet arrives at node $K + 1$)?

$$D =$$

- (c) [2 marks] Find the value of l that minimizes the end-to-end delay calculated in (b). (Hint: take the derivative of D w.r.t. l and then set $\frac{dD}{dl} = 0$ to obtain the critical value to minimize D .)

$$l =$$

This page is intentionally left blank.
Do **NOT** use it for your rough work.
Use it **ONLY** if you need extra space for your answer, in which case
please indicate the question number clearly.

This page is intentionally left blank.
Do **NOT** use it for your rough work.
Use it **ONLY** if you need extra space for your answer, in which case
please indicate the **question number clearly**.

=== END OF PAPER ===